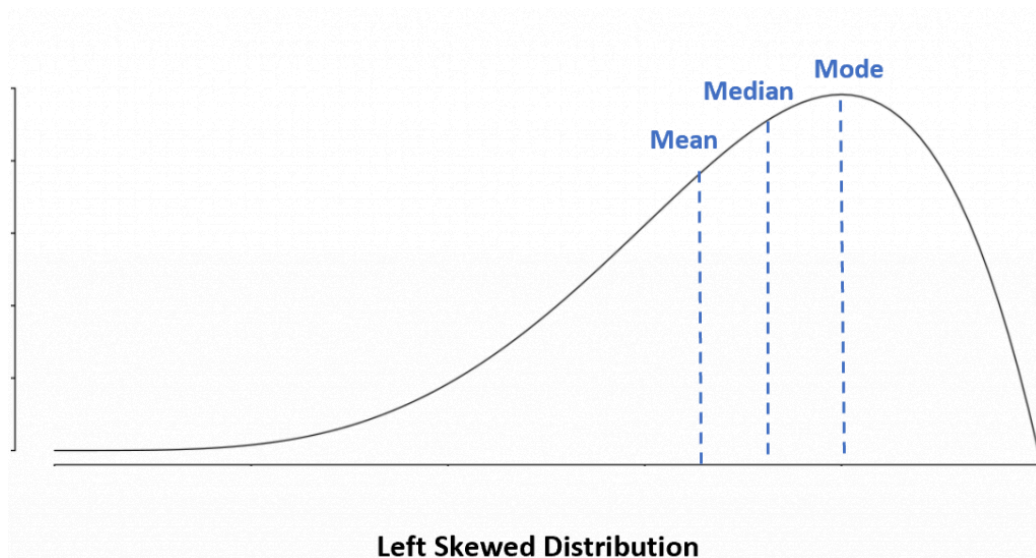


## 1. Item Pick Rate

- Table 1 is grouped by **item**, **month**, **zone**.

$$\text{Picks/Hr per SKU} = (\text{Picked Qty} / \text{Pickup Time (hrs)})$$
$$\text{Score} = (\text{Actual Rate} / \text{Target Rate}) \times 100$$

- Target Rate is found by finding the **centrality** of the Picks/Hr per SKU.



- However using **mean/average** won't be favourable because every warehouse dataset will be **skewed** because of wide variation on the higher end when high-velocity items are grouped together or multiple pickers work an aisle simultaneously.
- That's why we use the **median** of the dataset as target rate and find the score.

Finally zone pick rate is found by adding up all the SKU pick rate within each zone

### Threshold:

- Cumulative Item-Zone-Month observations: 183,474
- High-Performing Item-Zone Configurations (Score  $\geq 70$ ): 130,891 rows (71.3%)
- Underperforming Item-Zone Configurations (Score  $< 70$ ): 52,583 rows (28.7%)

**Output (first 20 rows):**

| itemCode    | pickMonth | storageSectionId | Item_Pick_Rate_Score | Threshold_70_Achieved |
|-------------|-----------|------------------|----------------------|-----------------------|
| 01463SE0911 | 2024-12   | AA               | 99.999774            | True                  |
| 0155305131  | 2024-11   | AA               | 99.999739            | True                  |
| 0155305751  | 2025-05   | AA               | 99.999887            | True                  |
| 0155305933  | 2024-05   | BB               | 100.000000           | True                  |
| 0155305933  | 2024-06   | BB               | 46.988050            | False                 |
| 0155305933  | 2024-07   | BB               | 27.986229            | False                 |
| 0155305933  | 2024-08   | BB               | 11.973980            | False                 |
| 0155305933  | 2024-09   | BB               | 14.395502            | False                 |
| 0155305933  | 2024-10   | BB               | 100.000000           | True                  |
| 0155305933  | 2025-01   | BB               | 100.000000           | True                  |
| 0155305933  | 2025-02   | BB               | 8.415871             | False                 |
| 0155305933  | 2025-03   | BB               | 100.000000           | True                  |
| 0155305933  | 2025-05   | BB               | 100.000000           | True                  |
| 0155305933  | 2025-07   | BB               | 82.000887            | True                  |
| 0155305933  | 2025-09   | BB               | 100.000000           | True                  |
| 0155309241  | 2024-11   | BB               | 100.000000           | True                  |
| 0155309241  | 2025-03   | BB               | 66.466273            | False                 |
| 0155309241  | 2025-07   | BB               | 100.000000           | True                  |
| 0155309241  | 2025-09   | BB               | 99.999741            | True                  |
| 0155309241  | 2025-10   | BB               | 32.419817            | False                 |

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## 2. SKU Throughput

- Table 1 is grouped by **item, month, zone**.

**SKU Throughput = Total Picked Units / Total Pickup Time**

**Threshold:**

>98th percentile for SKU Throughput

- Cumulative Item-Zone-Month observations: 183,474
- Elite-Speed Configurations (≥ 98th percentile): 3,670 rows (2.0%)
- Sub-Threshold Configurations (< 98th percentile): 179,804 rows (98.0%)

**Output:**

| itemCode    | pickMonth | storageSectionId | SKU_Throughput | Item_Throughput_Score | SKU_Speed_Pass |
|-------------|-----------|------------------|----------------|-----------------------|----------------|
| 01463SE0911 | 2024-12   | AA               | 0.012346       | 49.363595             | False          |
| 0155305131  | 2024-11   | AA               | 0.010676       | 42.687844             | False          |
| 0155305751  | 2025-05   | AA               | 0.024691       | 92.557758             | False          |
| 0155305933  | 2024-05   | BB               | 0.076922       | 100.000000            | True           |
| 0155305933  | 2024-06   | BB               | 0.010050       | 43.426855             | False          |
| 0155305933  | 2024-07   | BB               | 0.005984       | 25.501276             | False          |
| 0155305933  | 2024-08   | BB               | 0.002560       | 12.284043             | False          |
| 0155305933  | 2024-09   | BB               | 0.003079       | 12.463032             | False          |
| 0155305933  | 2024-10   | BB               | False          | 100.000000            | False          |
| 0155305933  | 2025-01   | BB               | 0.081395       | 100.000000            | True           |
| 0155305933  | 2025-02   | BB               | 0.001800       | 6.296337              | False          |
| 0155305933  | 2025-03   | BB               | 0.038095       | 100.000000            | True           |
| 0155305933  | 2025-05   | BB               | 0.057306       | 100.000000            | True           |

| itemCode   | pickMonth | storageSectionId | SKU_Throughput | Item_Throughput_Score | SKU_Speed_Pass |
|------------|-----------|------------------|----------------|-----------------------|----------------|
| 0155305933 | 2025-07   | BB               | 0.017544       | 73.654900             | False          |
| 0155305933 | 2025-09   | BB               | 0.090908       | 100.000000            | True           |
| 0155309241 | 2024-11   | BB               | 0.052910       | 100.000000            | True           |
| 0155309241 | 2025-03   | BB               | 0.007130       | 25.789732             | False          |
| 0155309241 | 2025-07   | BB               | 0.083332       | 100.000000            | True           |
| 0155309241 | 2025-09   | BB               | 0.010733       | 43.631901             | False          |
| 0155309241 | 2025-10   | BB               | 0.003478       | 15.297782             | False          |

---

### 3. Picker Performance

- Table 1 is grouped by **item, month, zone**.

$$\text{Per Picker: Score} = (\text{Actual Picks/Hr} / \text{Target Picks/Hr}) \times 100$$

- Similar to Item Pick Rate, we are finding the Target Picks/hr using **median**.
- Performance score of each picker is found and then grouped by item, zone and month (where **average picker performance** is calculated) to get per item picker performance score.

#### Threshold:

Cumulative Item-Zone-Month observations: 184,825

High-Performing Item-Zone Configurations (Score  $\geq 70$ ): 147,974 rows (80.1%)

Underperforming Item-Zone Configurations (Score  $< 70$ ): 36,851 rows (19.9%)

#### Output (first 20 rows):

| itemCode    | pickMonth | storageSectionId | n_pickers | Item_Picker_Performance_Score | Threshold_70_Achieved |
|-------------|-----------|------------------|-----------|-------------------------------|-----------------------|
| 01463SE0911 | 2024-12   | AA               | 1         | 56.322353                     | False                 |
| 0155305131  | 2024-11   | AA               | 1         | 100.000000                    | True                  |
| 0155305751  | 2025-05   | AA               | 1         | 100.000000                    | True                  |
| 0155305933  | 2024-05   | BB               | 1         | 100.000000                    | True                  |
| 0155305933  | 2024-06   | BB               | 1         | 48.626052                     | False                 |
| 0155305933  | 2024-07   | BB               | 1         | 53.514046                     | False                 |
| 0155305933  | 2024-08   | BB               | 1         | 19.590868                     | False                 |
| 0155305933  | 2024-09   | BB               | 1         | 35.590073                     | False                 |
| 0155305933  | 2024-10   | BB               | 1         | 100.000000                    | True                  |
| 0155305933  | 2025-01   | BB               | 1         | 100.000000                    | True                  |
| 0155305933  | 2025-02   | BB               | 1         | 30.051261                     | False                 |
| 0155305933  | 2025-03   | BB               | 1         | 100.000000                    | True                  |

| itemCode   | pickMonth | storageSectionId | n_pickers | Item_Picker_Performance_Score | Threshold_70_Achieved |
|------------|-----------|------------------|-----------|-------------------------------|-----------------------|
| 0155305933 | 2025-05   | BB               | 1         | 100.000000                    | True                  |
| 0155305933 | 2025-07   | BB               | 1         | 100.000000                    | True                  |
| 0155305933 | 2025-09   | BB               | 1         | 100.000000                    | True                  |
| 0155309241 | 2024-11   | BB               | 1         | 100.000000                    | True                  |
| 0155309241 | 2025-03   | BB               | 2         | 100.000000                    | True                  |
| 0155309241 | 2025-07   | BB               | 1         | 100.000000                    | True                  |
| 0155309241 | 2025-09   | BB               | 1         | 25.967200                     | False                 |
| 0155309241 | 2025-10   | BB               | 1         | 78.108134                     | True                  |

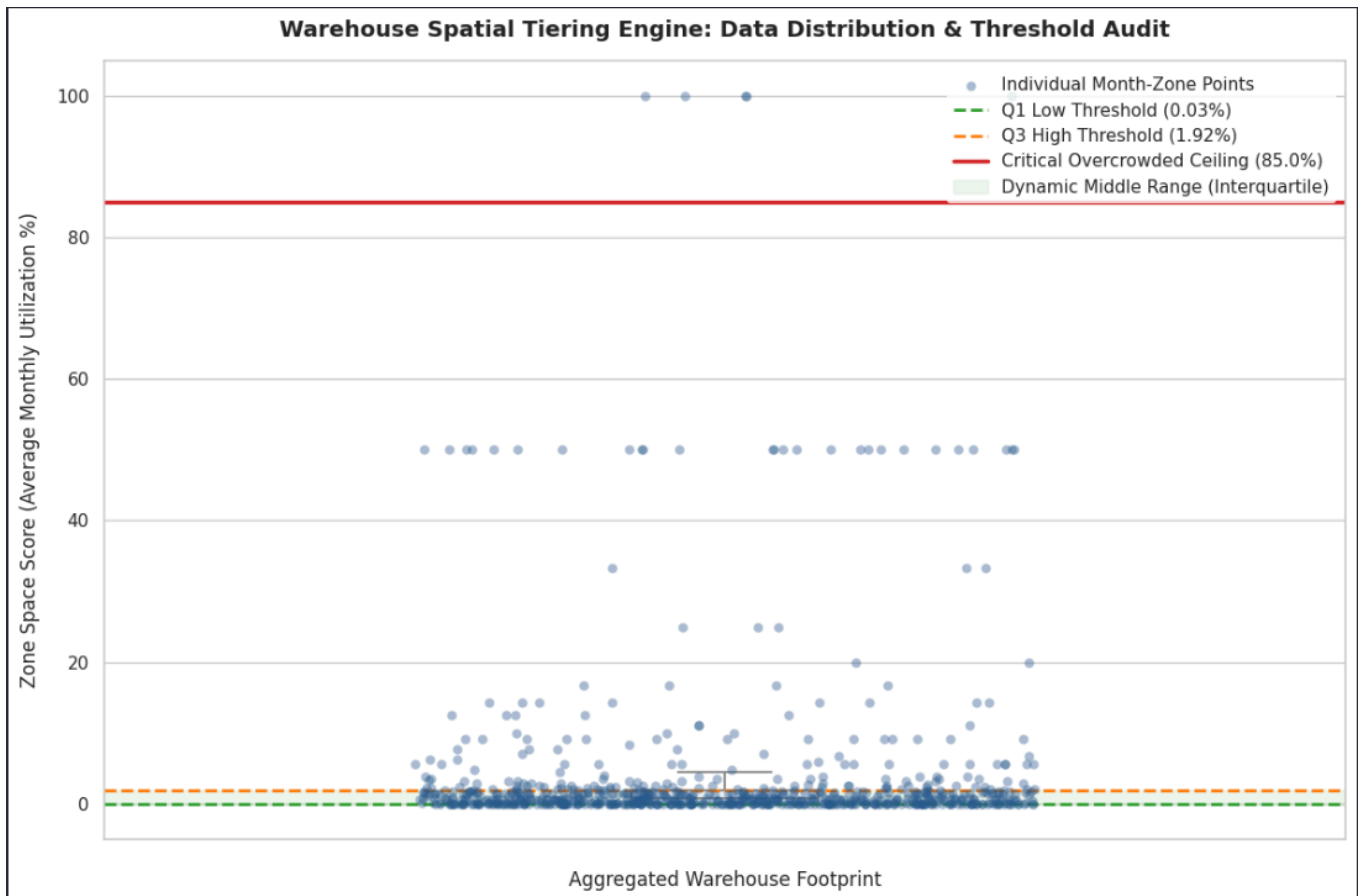
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## 4. Space Utilization

- Table 2 is grouped by item, month and zone.

$$\text{Space Score} = (\text{Occupied Locations} / \text{Total Locations}) \times 100$$

Ideal range: 75–85%



While the theoretical 'Ideal Range' for a healthy zone is 75%–85%, the graph reveals that **nearly 75% of our warehouse zones are operating at under 8.38% utilization**, with our median sitting right around 1.59%.

### Methodology:

- Aggregates: It calculates the mean utilization for every unique Zone-Month combination.

- Caps: The resulting Item\_Space\_Score is capped at 100.0%.
- Tiers:
  - If **score > 85%**, it is “Critically Overcrowded”.
  - If **score ≥ 75th percentile** (1.92%), it is a “High Activity Zone”.
  - If **score ≥ 25th percentile** (0.03%), it is a “Moderate Activity Zone”.
  - If **score < 25th percentile**, it is a “Low Activity Zone”.
- Broadcasts: Finally, it **maps this Zone-Month score back to every individual item located in that zone for that month**, ensuring item-level granularity for reporting.
- Total Broadcasted Item-Level Rows: 309,566
- Unique Zone-Month Combos Scored: 759
- Critically Overcrowded Zones (>85%): 5 rows
- High Activity Zones (≥ 1.92%): 188 rows
- Moderate Activity Zones (0.03% to 1.92%): 402 rows
- Low Activity Zones (< 0.03%): 164 rows

#### Methodology Adjustment:

- Because the warehouse is mostly empty, the dynamic thresholds (**instead of manual threshold**) have adjusted to reflect current operational realities. A zone with just 1.92% utilization is now flagged as 'High Activity.' This is a **relative classification**.
- By using dynamic percentiles (Q1=0.03%, Q3=1.92%) rather than fixed percentages, if the warehouse density increases, these thresholds will automatically scale upward; if it remains sparse, they stay sensitive to small variations.
- This gives us the zones which are available for relocating items.

#### Output (first 20 rows):

| itemCode   | month   | storageSectionId | Item_Space_Score |
|------------|---------|------------------|------------------|
| PF48       | 2024-03 | AA               | 0.02             |
| 9091902234 | 2024-03 | AA               | 0.02             |
| 534500W170 | 2024-03 | AB               | 0.02             |
| D2263C     | 2024-03 | AE               | 0.02             |
| MZ690563   | 2024-03 | AH               | 0.02             |



| itemCode    | month   | storageSectionId | Item_Space_Score |
|-------------|---------|------------------|------------------|
| 214-19A5    | 2024-03 | AA               | 0.02             |
| 135610S011  | 2024-03 | AB               | 0.02             |
| 485108Z190  | 2024-03 | AK               | 0.03             |
| 4351230310  | 2024-03 | AH               | 0.02             |
| 215-19C0-AL | 2024-03 | AA               | 0.02             |
| 215-19C0-A  | 2024-03 | AA               | 0.02             |
| GM389-A101L | 2024-03 | AJ               | 0.03             |
| 21996230    | 2024-03 | AF               | 0.02             |
| 84182733    | 2024-03 | AH               | 0.02             |
| 84182732    | 2024-03 | AH               | 0.02             |
| 22990034    | 2024-03 | AF               | 0.02             |
| 12696313    | 2024-03 | AH               | 0.02             |
| FR3Z16102A  | 2024-03 | AC               | 0.02             |
| 0437160070  | 2024-03 | AB               | 0.02             |
| 044650K580  | 2024-03 | BE               | 0.70             |

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## 5. Equipment Score

- Table 1 is grouped by **item, zone and month**.
- **1. Item Time Utilization Score**
  - Formula: (Active Equipment Time / 260 Total Shift Hours) × 100
- **2. Item Equipment Score (Relative Efficiency)**
  - Calculate **Picks per Equipment Hour**: Total Picks / Active Equipment Hours.
  - Determine **Specific Machine Baseline**: The median of "Picks per Equipment Hour" for that specific machine.
  - **Relative Efficiency**: (Actual Picks per Hour / Specific Machine Baseline) × 100.

**Output (first 20 rows):**

| itemCode    | pickMonth | storageSectionId | n_machines | Item_Time_Utilization_Score | Item_Equipment_Score |
|-------------|-----------|------------------|------------|-----------------------------|----------------------|
| 01463SE0911 | 2024-12   | AA               | 1          | 0.595406                    | 100.000000           |
| 0155305131  | 2024-11   | AA               | 1          | 12.882799                   | 85.412609            |
| 0155305751  | 2025-05   | AA               | 1          | 4.306410                    | 75.521343            |
| 0155305933  | 2024-05   | BB               | 1          | 5.254060                    | 100.000000           |
| 0155305933  | 2024-06   | BB               | 1          | 5.199786                    | 100.000000           |
| 0155305933  | 2024-07   | BB               | 1          | 3.873504                    | 100.000000           |
| 0155305933  | 2024-08   | BB               | 1          | 6.027350                    | 86.656673            |
| 0155305933  | 2024-09   | BB               | 1          | 4.356197                    | 100.000000           |
| 0155305933  | 2024-10   | BB               | 1          | 6.207158                    | 100.000000           |
| 0155305933  | 2025-01   | BB               | 1          | 5.723291                    | 100.000000           |
| 0155305933  | 2025-02   | BB               | 1          | 1.440171                    | 100.000000           |

| itemCode   | pickMonth | storageSectionId | n_machines | Item_Time_Utilization_Score | Item_Equipment_Score |
|------------|-----------|------------------|------------|-----------------------------|----------------------|
| 0155305933 | 2025-03   | BB               | 1          | 2.813248                    | 100.000000           |
| 0155305933 | 2025-05   | BB               | 1          | 9.831944                    | 100.000000           |
| 0155305933 | 2025-07   | BB               | 1          | 1.200962                    | 100.000000           |
| 0155305933 | 2025-09   | BB               | 1          | 10.359402                   | 100.000000           |
| 0155309241 | 2024-11   | BB               | 1          | 6.267628                    | 100.000000           |
| 0155309241 | 2025-03   | BB               | 2          | 1.038608                    | 100.000000           |
| 0155309241 | 2025-07   | BB               | 1          | 0.280556                    | 100.000000           |
| 0155309241 | 2025-09   | BB               | 1          | 1.491239                    | 100.000000           |
| 0155309241 | 2025-10   | BB               | 1          | 2.116346                    | 100.000000           |

Warehouse-wide average machine time utilization: **2.00%**

Warehouse-wide average machine relative efficiency: **89.39%**

- The data indicates that equipment remains largely idle, with an average time utilization of only 2.00%. However, when active, machines demonstrate high relative efficiency, averaging 89.39%.

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## 6. Putaway Location Summary

- Table 3 is grouped by item, zone and month.

$$\text{Putaway location accuracy} = (\text{Actual Bin} / \text{Proposed Bin}) * 100$$

Output:

| itemCode    | putawayMonth | storageSectionId | total_putaways | Item_Putaway_Accuracy_Score |
|-------------|--------------|------------------|----------------|-----------------------------|
| 0155304753  | 2024-04      | AA               | 1              | 0.0                         |
| 0155304753  | 2024-06      | AA               | 1              | 0.0                         |
| 0155305933  | 2024-09      | BB               | 1              | 100.0                       |
| 0155305933  | 2025-04      | BB               | 1              | 100.0                       |
| 0155309611  | 2024-09      | BB               | 1              | 100.0                       |
| 015532DR9A  | 2025-07      | LOC              | 1              | 0.0                         |
| 015532DR9A  | 2025-09      | BH               | 1              | 0.0                         |
| 0211704523  | 2024-08      | AD               | 1              | 0.0                         |
| 0211704523  | 2025-02      | AD               | 1              | 0.0                         |
| 0211707023  | 2025-04      | LOC              | 1              | 0.0                         |
| 02117070X3  | 2025-05      | AD               | 1              | 100.0                       |
| 0211724523  | 2024-09      | AC               | 1              | 100.0                       |
| 02117245XC  | 2024-11      | AD               | 1              | 0.0                         |
| 02117880XC  | 2024-03      | AD               | 1              | 0.0                         |
| 02117880XC  | 2024-11      | AD               | 1              | 100.0                       |
| 0211791023  | 2024-11      | AD               | 1              | 0.0                         |
| 022510-4170 | 2025-04      | AH               | 1              | 100.0                       |
| 022510-4170 | 2025-07      | BB               | 1              | 100.0                       |
| 041110D541  | 2024-05      | BA               | 1              | 0.0                         |
| 041110S023  | 2025-03      | LOC              | 1              | 0.0                         |

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## 7. Dock to Stock Time

- Table 3 is grouped by **item, zone and month**

**Dock to Stock time = Time difference between unloading time and Putaway time per Zone per month**

- The dataset has already got the time difference for each item in the “totalTimeInSec” column. So we have to multiply it by the quantity to get time for all the items.
- Finally divide the dock to stock time with the volume of the item to get the **weighted dock to stock time**.
- When we group by these zone-month values along with item, we get the item’s dock to stock time.

### Output:

| itemCode   | putawayMonth | storageSectionId | weighted_d2s_seconds | Item_D2S_Score |
|------------|--------------|------------------|----------------------|----------------|
| 0155304753 | 2024-04      | AA               | 432.999869           | 47.023361      |
| 0155304753 | 2024-06      | AA               | 301.999497           | 67.626800      |
| 0155305933 | 2024-09      | BB               | 1840.999080          | 45.500849      |
| 0155305933 | 2025-04      | BB               | 3307.998346          | 5.834684       |
| 0155309611 | 2024-09      | BB               | 1606.998393          | 49.660441      |
| 015532DR9A | 2025-07      | LOC              | 508.999491           | 63.457207      |
| 015532DR9A | 2025-09      | BH               | 920.999079           | 60.704356      |
| 0211704523 | 2024-08      | AD               | 1576.984230          | 51.791277      |
| 0211704523 | 2025-02      | AD               | 3263.967360          | 5.381818       |
| 0211707023 | 2025-04      | LOC              | 420.995790           | 62.182604      |

| itemCode    | putawayMonth | storageSectionId | weighted_d2s_seconds | Item_D2S_Score |
|-------------|--------------|------------------|----------------------|----------------|
| 02117070X3  | 2025-05      | AD               | 2940.970590          | 7.902555       |
| 0211724523  | 2024-09      | AC               | 2074.989625          | 40.025467      |
| 02117245XC  | 2024-11      | AD               | 1140.988590          | 49.349845      |
| 02117880XC  | 2024-03      | AD               | 49.999500            | 57.095662      |
| 02117880XC  | 2024-11      | AD               | 750.992490           | 57.461300      |
| 0211791023  | 2024-11      | AD               | 2079.979200          | 31.021672      |
| 022510-4170 | 2025-04      | AH               | 2741.972580          | 16.153431      |
| 022510-4170 | 2025-07      | BB               | 1023.997952          | 53.096847      |
| 041110D541  | 2024-05      | BA               | 75.999240            | 74.104143      |
| 041110S023  | 2025-03      | LOC              | 65.999340            | 78.759640      |

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## 8. Putaway Throughput

- Table 3 is grouped by **item, zone and month**.

$$\text{Throughput} = \Sigma(\text{putaway qty}) \div \Sigma(\text{lead time})$$

- We find the monthly **75th percentile** velocity as the **throughput ceiling** for calculations.
- Raw throughput / throughput ceiling = Item Throughput Score

### Output:

| itemCode   | putawayMonth | storageSectionId | raw_throughput_rate | Item_Putaway_Throughput_Score |
|------------|--------------|------------------|---------------------|-------------------------------|
| 0155304753 | 2024-04      | AA               | 0.076212            | 100.000000                    |
| 0155304753 | 2024-06      | AA               | 0.019868            | 100.000000                    |
| 0155305933 | 2024-09      | BB               | 0.010864            | 100.000000                    |
| 0155305933 | 2025-04      | BB               | 0.006046            | 40.480741                     |
| 0155309611 | 2024-09      | BB               | 0.006223            | 94.054697                     |
| 015532DR9A | 2025-07      | LOC              | 0.019646            | 100.000000                    |
| 015532DR9A | 2025-09      | BH               | 0.010858            | 100.000000                    |
| 0211704523 | 2024-08      | AD               | 0.000634            | 8.621712                      |
| 0211704523 | 2025-02      | AD               | 0.000306            | 1.500491                      |
| 0211707023 | 2025-04      | LOC              | 0.002375            | 15.903835                     |
| 02117070X3 | 2025-05      | AD               | 0.000340            | 1.767187                      |
| 0211724523 | 2024-09      | AC               | 0.000964            | 14.568279                     |
| 02117245XC | 2024-11      | AD               | 0.000876            | 6.011528                      |
| 02117880XC | 2024-03      | AD               | 0.020000            | 34.993879                     |
| 02117880XC | 2024-11      | AD               | 0.001332            | 9.133359                      |

| itemCode    | putawayMonth | storageSectionId | raw_throughput_rate | Item_Putaway_Throughput_Score |
|-------------|--------------|------------------|---------------------|-------------------------------|
| 0211791023  | 2024-11      | AD               | 0.000481            | 3.297670                      |
| 022510-4170 | 2025-04      | AH               | 0.000365            | 2.441836                      |
| 022510-4170 | 2025-07      | BB               | 0.004883            | 44.990846                     |
| 041110D541  | 2024-05      | BA               | 0.013158            | 65.756600                     |
| 041110S023  | 2025-03      | LOC              | 0.015152            | 90.854580                     |

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## Weight Finding Methods:

### 1. CRITIC Method

It finds the weights for parameters purely mathematically using these two methods:

- **Standard Deviation:** Scores based on how much the values vary from item to item. High Deviation = Higher Weight and vice versa.
- **Correlation:** If two parameters have high correlation, then the weights are reduced and balanced among themselves to avoid double counting.

#### Output:

- Item\_Pick\_Rate\_Score: 14.43%
  - Item\_Throughput\_Score: 12.93%
  - Item\_Picker\_Performance\_Score: 12.85%
  - Item\_Space\_Score: 1.99%
  - Item\_Equipment\_Score: 8.54%
  - Item\_Putaway\_Throughput\_Score: 15.23%
  - Item\_D2S\_Score: 10.28%
  - Item\_Putaway\_Accuracy\_Score: 23.74%
- 

### 2. Group AHP

- A table with all the parameters as rows and columns.
- Experts score each parameter, comparing how important is one parameter more than another.
- These comparisons are then used to find the weights of each parameter.

#### Output:

- Item\_Pick\_Rate\_Score: 18.42%
- Item\_Throughput\_Score: 20.35%

- Item\_Picker\_Performance\_Score: 5.45%
  - Item\_Space\_Score: 9.86%
  - Item\_Equipment\_Score: 2.87%
  - Item\_Putaway\_Throughput\_Score: 18.77%
  - Item\_D2S\_Score: 5.50%
  - Item\_Putaway\_Accuracy\_Score: 18.78%
- 

### 3. Hybrid Score:

- This method includes both **CRITIC** and **Group AHP**.
- Hybrid Score =  $(0.5) * \text{CRITIC} + (0.5) * \text{Group AHP}$

#### **Output:**

- Item\_Pick\_Rate\_Score: 16.42%
  - Item\_Throughput\_Score: 16.64%
  - Item\_Picker\_Performance\_Score: 9.15%
  - Item\_Space\_Score: 5.93%
  - Item\_Equipment\_Score: 5.71%
  - Item\_Putaway\_Throughput\_Score: 17.00%
  - Item\_D2S\_Score: 7.89%
  - Item\_Putaway\_Accuracy\_Score: 21.26%
-

## Zone Efficiency & TOPSIS

### How are we finding it:

- All 8 parameters are **normalized** to the same range (0.0 - 1.0) and merged to a single dataframe.
  - This dataframe can be used by a ML model to find the zone efficiency.
- 

### Trying to find the ML model:

- The type of dataset we have has **only independent variables**, which eliminates any Supervised ML approach (like multiple linear regression), because there is **no pre-defined Zone Efficiency Score column** we can use to **train** the model.
  - This kind of dataset is **applicable for Unsupervised ML approach**, where we can use K-Means to cluster zones into different efficiencies (High, Moderate, Low).
  - Among Unsupervised ML approaches, the **K-Means Model** is the most optimal for determining the zone. Other models like GMM give a mixture of scores for each zone.
  - K-Means considers all parameters equally, it **doesn't have any weights** for each of the parameters. This is because K-Means **plots the data geometrically** and finds the distance between them for clustering.
  - As some of the parameters have less importance/weightage than others, we can use **Weighted K-Means Model** for assigning different weights to each parameter and forming clusters.
- 

### How Zone Efficiency Score is calculated:

- **Weighted K-Means only groups data**, it does not give a smooth 0-100 score. To generate a continuous ranking, we use **TOPSIS**.

- **TOPSIS** - the optimal item-zone-month profile should have the **shortest geometric distance to the Ideal Solution** and the **farthest distance from the Anti-Ideal Solution**. High-Efficiency Center (t1) = **Ideal Solution**, while the Low-Efficiency Center (t3) = **Anti-Ideal Solution**.
- By using the **kmeans.transform()** function, the model determines the **linear geometric distances** (represented as d<sub>1</sub>, d<sub>2</sub>, and d<sub>3</sub>) between individual data points and these established **anchor centers**.

**Tier 3 Assignment (Low Efficiency):** The score is mapped within the **0 to 33.33** bracket. Following TOPSIS logic, as its distance to the nightmare anchor (**d3**) increases relative to the moderate anchor (**d2**), its score climbs closer to **33.33**:

$$\text{Score} = 33.33 \times (1 - (d_3 / (d_3 + d_2 + 1e^{-5})))$$

**Tier 2 Assignment (Moderate Efficiency):** The score occupies the median **33.33 to 66.66** range. The calculation identifies if the zone **skews** toward the lower tier (d<sub>3</sub> < d<sub>1</sub>) or the higher tier to apply a normalized fractional step:

$$\text{Score (leaning low)} = 33.33 + (16.67 \times (d_3 / (d_2 + d_3 + 1e^{-5})))$$

$$\text{Score (leaning high)} = 50.00 + (16.66 \times (1 - (d_2 / (d_1 + d_2 + 1e^{-5}))))$$

**Tier 1 Assignment (High Efficiency):** The score enters the **66.66 to 100.0** elite bracket. It evaluates how close the zone is to peak performance (**t1**) relative to the moderate center (**t2**). As the distance to the ideal dream anchor (**d1**) approaches zero, the score scales tightly up to a perfect **100.0**:

$$\text{Score} = 66.66 + (33.34 \times (d_2 / (d_1 + d_2 + 1e^{-5})))$$

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**Output (first 20 rows):**

#### **CRITIC (Data-Driven) Tier Distribution:**

- Tier 1 (High Efficiency): 192,911 rows (62.12%)
- Tier 3 (Low Efficiency): 61,266 rows (19.73%)
- Tier 2 (Moderate Efficiency): 56,359 rows (18.15%)

### AHP (Expert Strategy) Tier Distribution:

- Tier 1 (High Efficiency): 197,480 rows (63.59%)
- Tier 2 (Moderate Efficiency): 62,086 rows (19.99%)
- Tier 3 (Low Efficiency): 50,970 rows (16.41%)

### Hybrid (Combined) Tier Distribution:

- Tier 1 (High Efficiency): 198,420 rows (63.9%)
- Tier 3 (Low Efficiency): 61,622 rows (19.84%)
- Tier 2 (Moderate Efficiency): 50,494 rows (16.26%)

| month   | storageSectionId | itemCode    | Data_Efficiency_Score | Strategy_Efficiency_Score | Hybrid_Efficiency_Score |
|---------|------------------|-------------|-----------------------|---------------------------|-------------------------|
| 2024-03 | AA               | 01463SE0911 | 87.853462             | 92.951267                 | 90.188287               |
| 2024-03 | AA               | 0155305933  | 93.975353             | 95.473213                 | 93.975991               |
| 2024-03 | AA               | 0155308231  | 93.975353             | 95.473213                 | 93.975991               |
| 2024-03 | AA               | 0155309241  | 93.975353             | 95.473213                 | 93.975991               |
| 2024-03 | AA               | 0155309611  | 93.975353             | 95.473213                 | 93.975991               |
| 2024-03 | AA               | 015532DR9A  | 93.975353             | 95.473213                 | 93.975991               |
| 2024-03 | AA               | 041110C098  | 94.482730             | 95.619373                 | 94.262029               |
| 2024-03 | AA               | 0431112110  | 23.364709             | 60.613448                 | 22.919732               |
| 2024-03 | AA               | 0431136082  | 21.852706             | 60.016773                 | 21.579078               |
| 2024-03 | AA               | 0431353010  | 23.457269             | 60.409082                 | 23.163005               |
| 2024-03 | AA               | 0442842101  | 24.262439             | 61.266212                 | 24.151391               |
| 2024-03 | AA               | 0444548040  | 24.553332             | 61.395848                 | 24.486769               |
| 2024-03 | AA               | 0447850040  | 24.274661             | 61.270954                 | 24.162121               |
| 2024-03 | AA               | 0447860050  | 24.433581             | 61.374101                 | 24.337938               |

| month   | storageSectionId | itemCode   | Data_Efficiency_Score | Strategy_Efficiency_Score | Hybrid_Efficiency_Score |
|---------|------------------|------------|-----------------------|---------------------------|-------------------------|
| 2024-03 | AA               | 0447860080 | 24.456948             | 61.382717                 | 24.358011               |
| 2024-03 | AA               | 0447922020 | 87.863216             | 87.617226                 | 87.657054               |
| 2024-03 | AA               | 0447933050 | 87.723831             | 87.610821                 | 87.612913               |
| 2024-03 | AA               | 0447960030 | 93.975353             | 95.473213                 | 93.975991               |
| 2024-03 | AA               | 0447960270 | 93.975353             | 95.473213                 | 93.975991               |
| 2024-03 | AA               | 0449537034 | 93.975353             | 95.473213                 | 93.975991               |

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